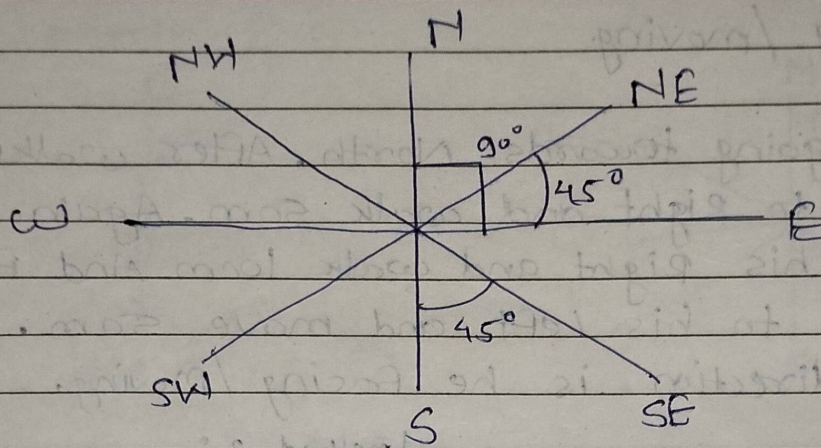


2/4/2025

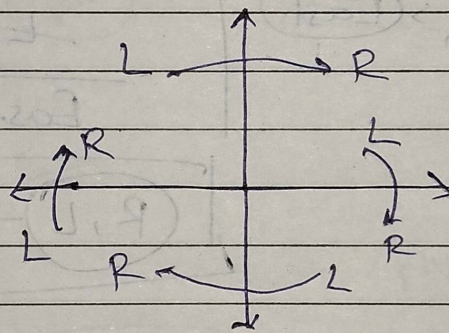
29. Direction and Distance

PAGE NO.:

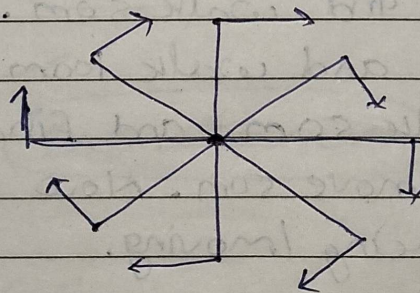
DATE: / /



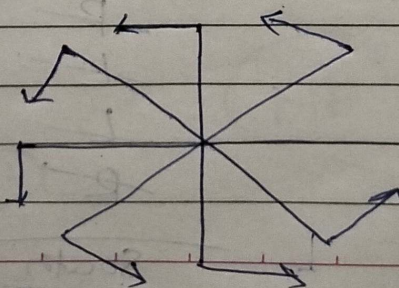
Turn Right
Turn Left



Right $\rightarrow$ 90° clockwise



Turn Left $\rightarrow$ 90° Anticlockwise

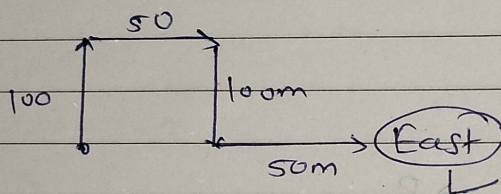


TYPE-1:- Facing /moving

- ① A person going towards North. After walking 100m he turns to Right and walk 50m. Again he turns to his Right and walk 100m And Finally he turns to his Left and move 50m. Now in which direction is he facing /moving.



method-1:-



method-2:-

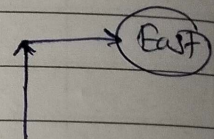
North

R

R

L

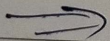
East



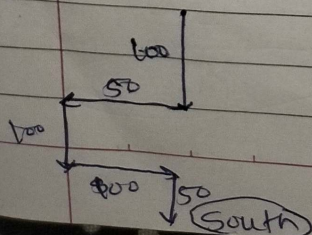
(R, L)

→ No change

- ② A Person going towards south. After walking 100m he turns to Right and walk 50m. Again he turns to his Left and walk 100m and Again turns Left and walk 50m and finally he turns to his Right and move 50m. Now in which direction is he facing /moving.



(M1)



(M2)

South

R

L

L

R

South

R, L → No change

① R, L $\rightarrow$ No change
 ② 2R / 2L $\xrightarrow{180^\circ}$ opposite

③ 3R $\xrightarrow{270^\circ}$ 1L

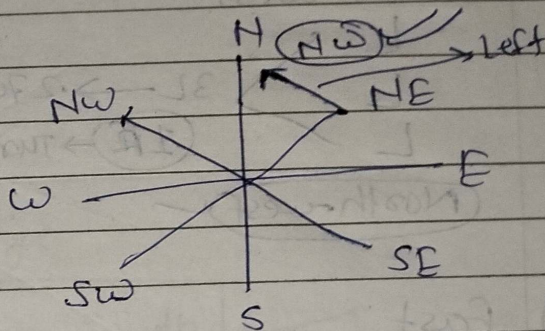
④ 3L $\xrightarrow{270^\circ}$ 1R

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3) North-East

R
 L
 L
 R
 L

North-west



4) North

R
 R
 R

180°

North opposite South

L
 South

5) East

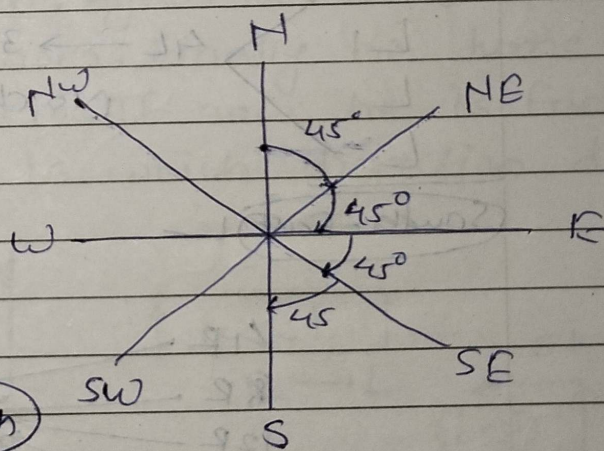
R
 R
 R

3R $\rightarrow$ 270°

1L move

North

East Left North



6) North-East

L
 L $\xrightarrow{180^\circ}$ opposite

R
 1L

NE opposite SW

South-west

5) $4R/4L \rightarrow$ No change (360°)

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7) South-west

L
L
L $\rightarrow$ $3L \rightarrow 270^\circ$
 $(1R) \rightarrow$ Turn

North-west

8) East

R
R
R
R $\rightarrow$ $4R \rightarrow 360^\circ$
No change

East

9) South-East

L
L
L
L $\rightarrow$ $4L \rightarrow 360^\circ$
No change

South-East

$4R$
 $8R$
 $12R$
 $16R$
 $40R$
 $100R$
 $300R$
No change

10) North

$300R$ $100R$
 $200L$
 $\rightarrow$ Multiple of 4 so
No change

North

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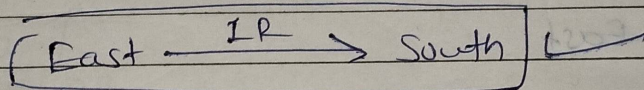
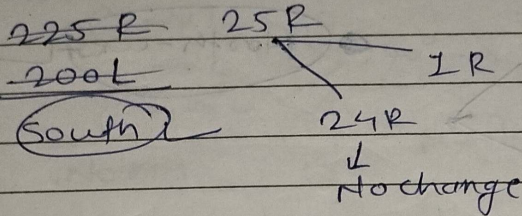
① Type-I :-
start → ?

② Type-II
? → End

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ii) East



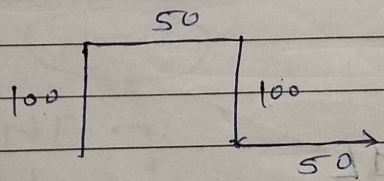
iii) Type-II

? → End

① A Person turns to his Right, after walking 100m straight and walk 50m. Again he turns to his Right and walk 100m. Finally he turns to his left and walk 50m. IF now he is facing towards north, then is which direction did he started?



method-1



N → West
East → North

West

R → L

R

L

North

Left

West

N

Reverse direction

R → L and L → R Change

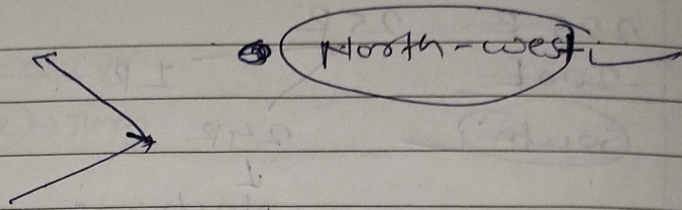
direction

2)

NW

R $\rightarrow$ L~~L~~~~R~~~~R~~~~L~~

North-East



3)

NW

~~L~~~~R~~R $\rightarrow$ LR $\rightarrow$ L180° $\rightarrow$ oppositeSE $\xrightarrow{\text{opp}}$ NW

North west

South-East

4)

SW

R $\rightarrow$ LR $\rightarrow$ LR $\rightarrow$ LR $\rightarrow$ L360° $\rightarrow$ No change

South-west

South-west

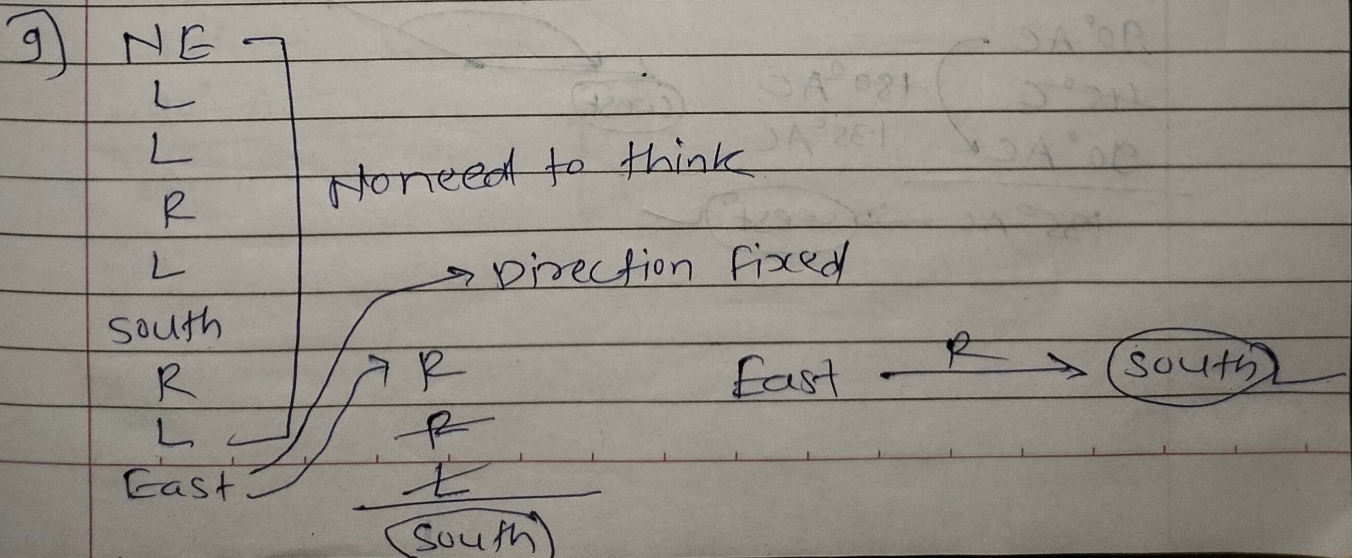
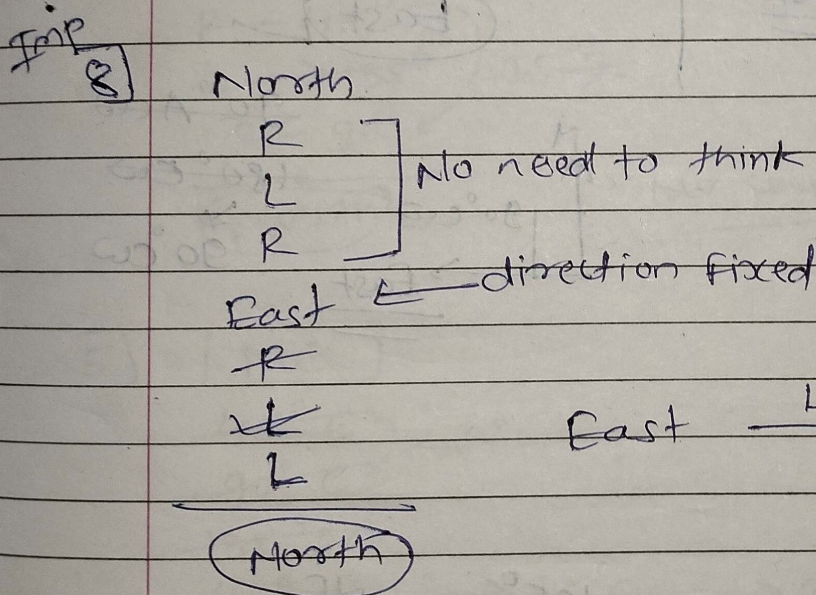
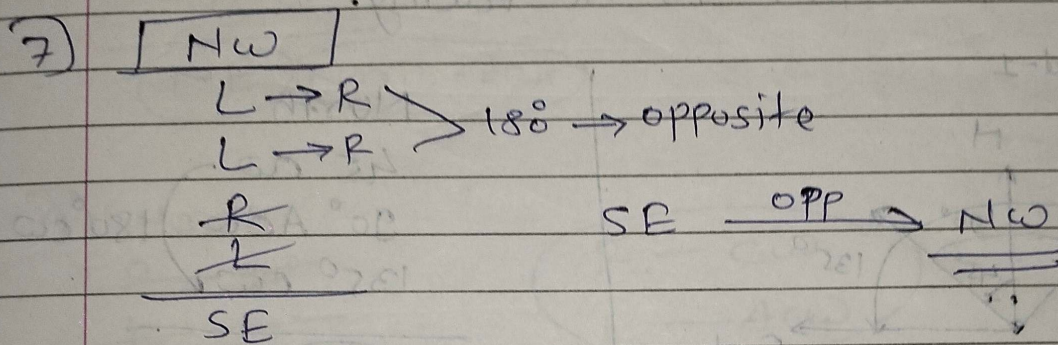
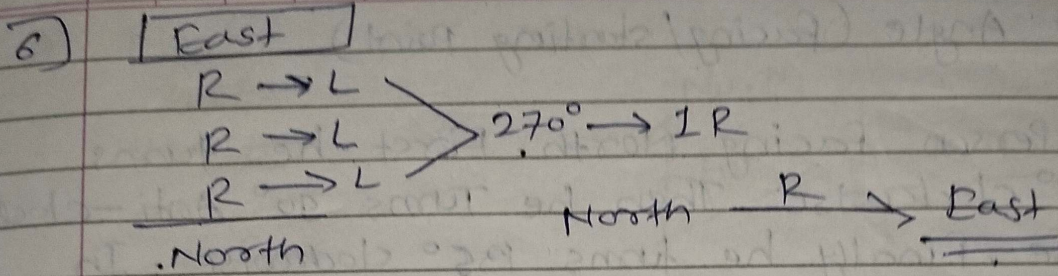
5)

South

L $\rightarrow$ RL $\rightarrow$ RL $\rightarrow$ R270° $\rightarrow$ 1L

west

west $\xrightarrow{L}$ south



① (CW, ACW) $\rightarrow$ No change

45° CW
45° ACW $\rightarrow$ No change

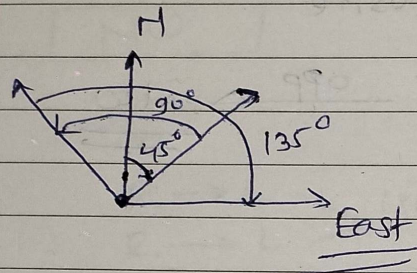
PAGE NO.:

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TYPE-III: Angle (facing/starting point)

① A Person facing North. First he turns 45° clockwise. Then he turns 90° Anti-clockwise. Finally he turns 135° clockwise. In which direction is he facing.

$\Rightarrow$ method-1



North

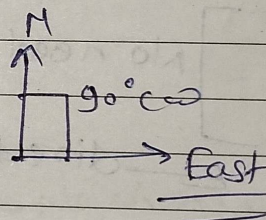
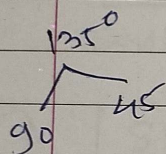
45° CW

90° ACW

135° CW

180° CW

East



90° ACW

180° CW

90° CW

② NB

45° C

90° AC

45° C

90° AC

45° C

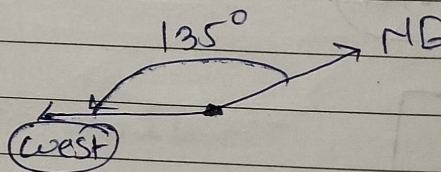
90° AC

180° AC

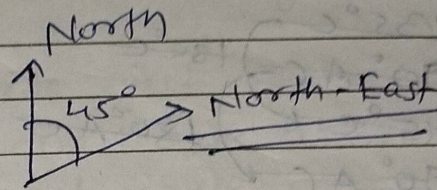
135° AC

135° AC

West



- 3) North
 $45^\circ E$
 $135^\circ AC$
 $45^\circ AC$
 $135^\circ C$
 $45^\circ C$



NE North-East

- 4) North
 $R - 90^\circ$
 $L - 90^\circ$
 $R - 45^\circ$
 $L \rightarrow 135^\circ$
 $R - 90^\circ$

$R - CW$
 $L - ACW$

North

No change

5) NE

- $45^\circ C$
 $90^\circ C$
 $135^\circ AC$

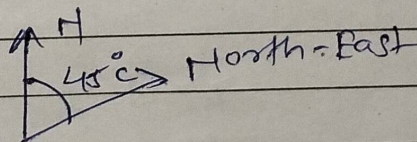
$45^\circ AC$

$45^\circ AC \rightarrow 45^\circ C \rightarrow$ reverse direction

$90^\circ C$

$90^\circ A$

North



6)

NE

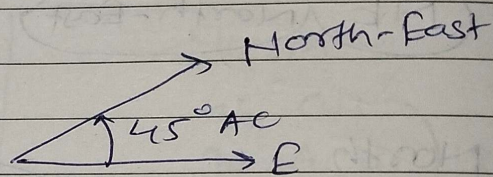
135°C
 90°AC
 45°C
 90°AC

180°C
 180°AC

$135^\circ \text{C} \rightarrow 45^\circ \text{C} \rightarrow 45^\circ \text{AC} \rightarrow \text{reverse}$

45°AC
 $45^\circ \text{AC} \rightarrow 90^\circ \text{AC}$

East



(T1)

Start

→ ?

(T2)

?

→ End

(T3) Angle

(1) Start

→ ?

(2) ?

→ End

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Type-4 :- Direction & distance from starting point (Diagram needed)

① A man is going towards east. After walking 100m he turns left and walk 25m. Again he turns to his left and walk 45m. Finally he turns his left and walk 25m then find

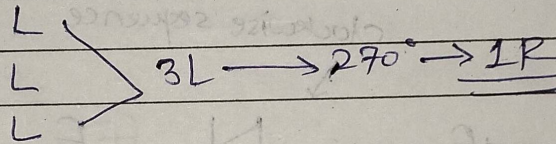
① In which direction is he Facing

② Find his direction & distance from starting point

→

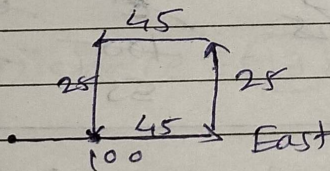
① Facing :-

East



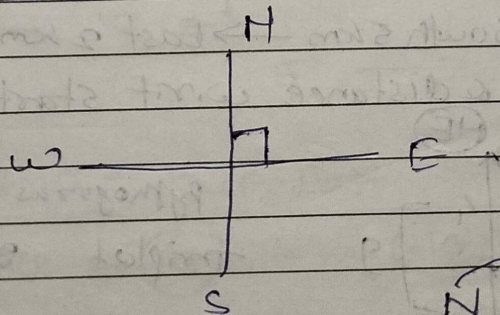
~~North~~ South

②



East
55m

$$100 - 45 = 55m$$



~~N-K~~

~~E-AE~~

~~N-S~~

~~W-E~~

Pythagoras theorem
90° 90° 90°

$$h^2 = p^2 + b^2$$

Triplets :-

(1) 5, 4, 3

(2) 10, 8, 6

(3) 15, 12, 9

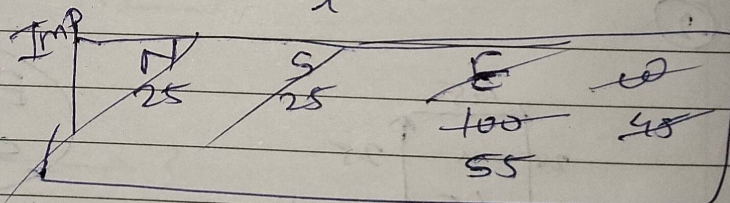
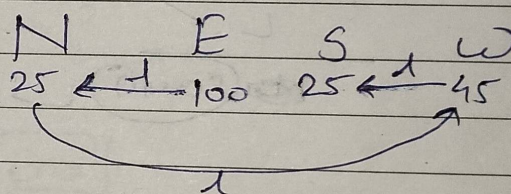
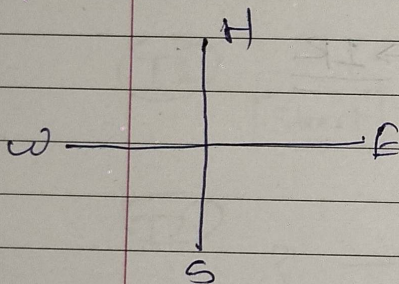
(4) 20, 16, 12

(5) 13, 12, 5

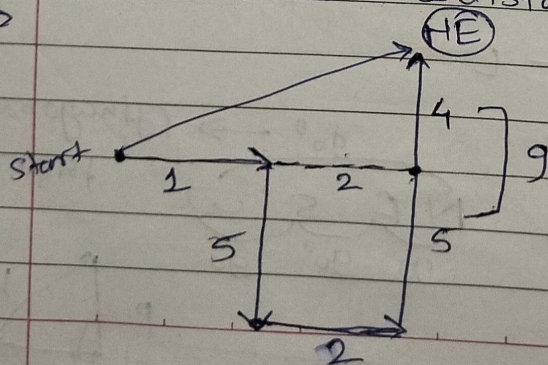
Without Diagram ans:-

Imp

Right → move forward

← left move Backward
clockwise sequenceEast → 55m

(2) East 1 km → South 5 km → East 2 km → North 9 km
Find his direction & distance w.r.t starting Point.



Pythagoras theorem

Triplet ~~5, 4, 3~~ 5, 4, 3

distance = 5

direction = North-East

(M2)

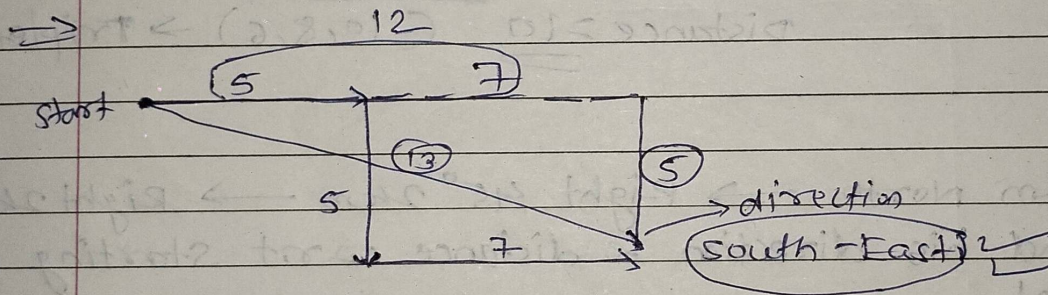
N	E	5	3
9	1	5	
4	<u>2</u>		
	3		

Direction = North - East

Distance = 5

(5, 4, 3) → Triplet.

- (3) 5km East → 5km South → left 7km
Find his direction & distance w.r.t starting point.



Distance = (13) → triplet (13, 12, 5)

(M2)

N	E	S	5
	5	5	
	7	←	
	<u>12</u>	5	

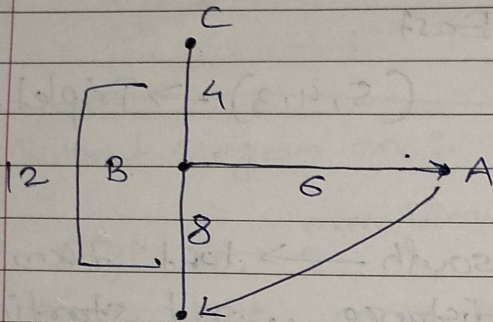
Pythagoras Use
Successive direction

Direction = South-East

Distance = (13) → (13, 12, 5) → Triplet

- (4) A is 6km East to B. C is 4km North to B. D is 12km South to C. Find direction & distance of D w.r.t A.

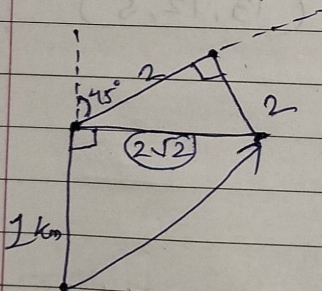
⇒ Without diagram can't be solve these type question



Direction = South-west

Distance = 10 (10, 8, 6) → Triplet

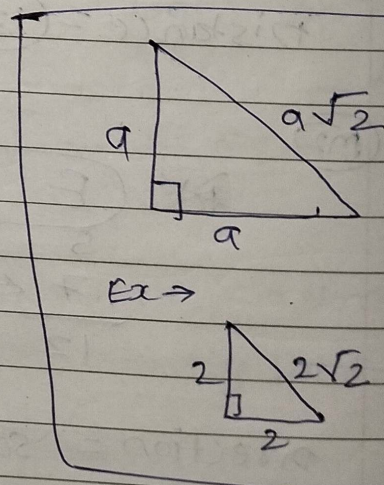
- (5) 1km North → Right 45° 2km → Right 2km
Find his direction & distance w.r.t starting point.



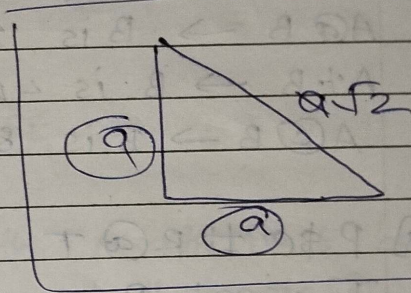
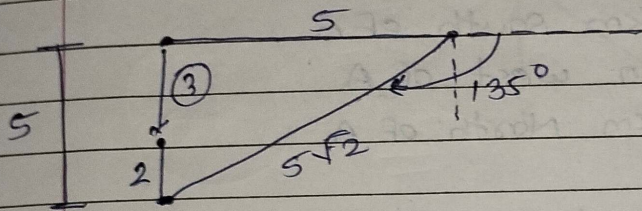
Direction = North-East

$$\begin{aligned} \text{Distance} &= \sqrt{1^2 + (2\sqrt{2})^2} \\ &= \sqrt{1 + 4 \times 2} \\ &= \sqrt{1 + 8} \\ &= \sqrt{9} \end{aligned}$$

Distance = 3



- ⑥ 5 km East $\rightarrow$ Right 135° $(5\sqrt{2})$ $\rightarrow$ North 2 km
Find direction & distance w.r.t starting point.
 $\Rightarrow$



Direction = South

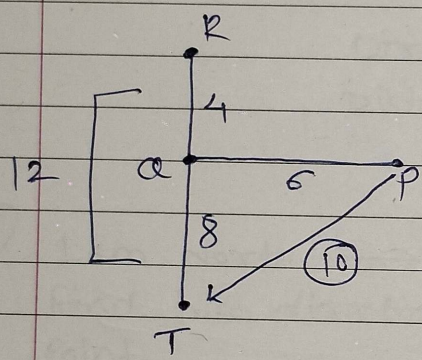
Distance = 3 km

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~~Find~~ Coded direction

- ① A\$B → A is 6km East of B
A#B → B is 4km North of A
A@B → B is 12km South of A
A% B → B is 4km West of A
A©B → B is 8km North of A.

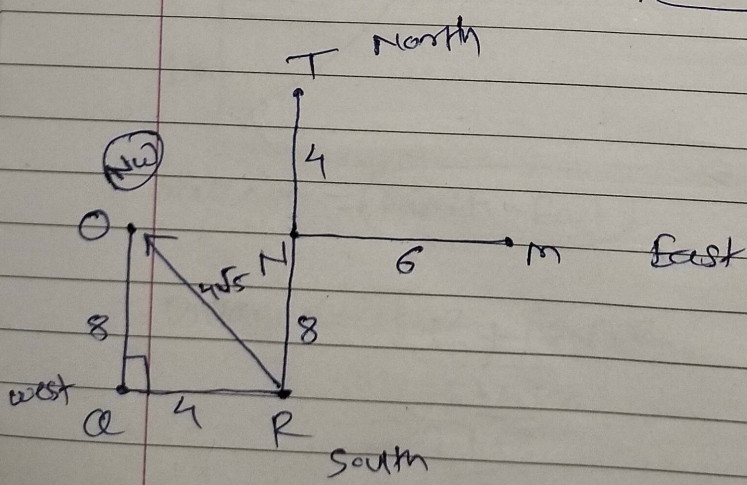
a) P\$Q#R@T Find direction and distance of T w.r.t P.



Direction = South-west

Distance = 10
— (10, 8, 6)
Triplet

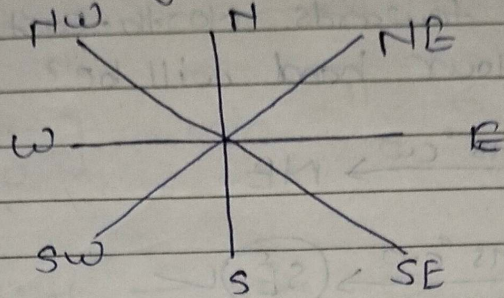
b) M\$N, N#T, T@R, R%Q©O, Find direction and distance of O w.r.t R



Direction = North-west

Distance = $\sqrt{8^2 + 4^2}$
= $\sqrt{64 + 16}$
= $\sqrt{80}$
= $\sqrt{16 \times 5}$
Distance = $4\sqrt{5}$

Type-5:- Direction change



- ① IF South East become North, North East become west and so on then west becomes ?

⇒

South East $\xrightarrow{135^\circ \text{ACW}}$ North

west $\xrightarrow{135^\circ \text{ACW}}$ South-East ✓

- ② IF North $\longrightarrow$ South-East then
South $\longrightarrow$?

⇒

North $\xrightarrow{135^\circ \text{CW}}$ South-East

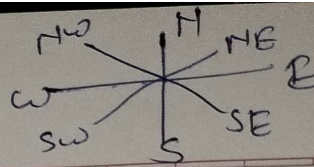
South $\xrightarrow{135^\circ \text{CW}}$ North-west ✓

- ③ IF South-East $\longrightarrow$ South-west then
South $\longrightarrow$?

⇒

South-East $\xrightarrow{90^\circ \text{CW}}$ South-west

South $\xrightarrow{90^\circ \text{CW}}$ west ✓



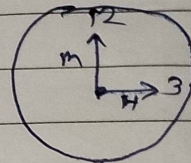
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- 4) It is 3:00 o'clock. If at this time minute hand points towards North-East, then in which direction Hour hand will be?

min (N) $\xrightarrow{45^\circ \text{ CW}}$ NE

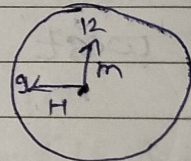
Hr (E) $\xrightarrow{45^\circ \text{ CW}}$ SE



- 5) It is 9:00 o'clock. min $\longrightarrow$ NW and Hr $\longrightarrow$?

min (N) $\xrightarrow{45^\circ \text{ ACW}}$ NW

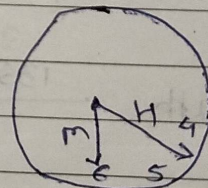
Hr (W) $\xrightarrow{45^\circ \text{ ACW}}$ SW



- 6) It is 4:30 and Hr $\longrightarrow$ SW then min $\longrightarrow$?

Hr (SE) $\xrightarrow{90^\circ \text{ CW}}$ SW

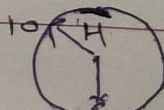
min (S) $\xrightarrow{90^\circ \text{ CW}}$ West



- 7) 10:30, Hr $\longrightarrow$ North
1:30 Hr $\longrightarrow$?

10:30 Hr (NW) $\xrightarrow{45^\circ \text{ CW}}$ North

1:30 Hr (NE) $\xrightarrow{45^\circ \text{ CW}}$ East



8) 2:30, MH $\longrightarrow$ west
 4:30, MH $\longrightarrow$?

$\Rightarrow$ 2:30] — min तो 6 पे है होता है
 4:30]

50,

2:30 MH $\longrightarrow$ west

4:30 MH $\longrightarrow$ (west)

9) 1:00, MH $\longrightarrow$ SW
 4:00, MH $\longrightarrow$?

$\Rightarrow$ 1:00] — min तो 12 पे है होता है
 4:00]

50,

1:00 MH $\longrightarrow$ SW

4:00 MH $\longrightarrow$ (SW)

Type-6:-

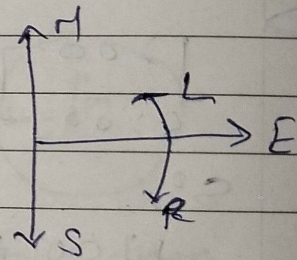
- ① A man performing yoga with Head down if at this time he is facing towards west then in which direction his left hand be:



Head down आलं की सीधा करे दो because
 only face change left, right not change.

Head down Face $\longrightarrow$ west $\xrightarrow{\text{opposite}}$ East

LH $\longrightarrow$ North
 RH $\longrightarrow$ South

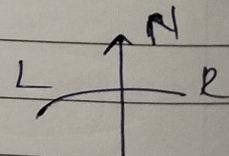


- ② Head down yoga
 Face $\longrightarrow$ South
 LH = ? , RH = ?

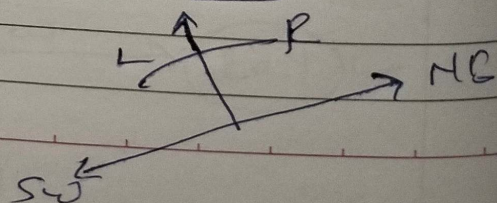


Head down Face $\longrightarrow$ South $\xrightarrow{\text{opposite}}$ North

LH $\longrightarrow$ west
 RH $\longrightarrow$ East



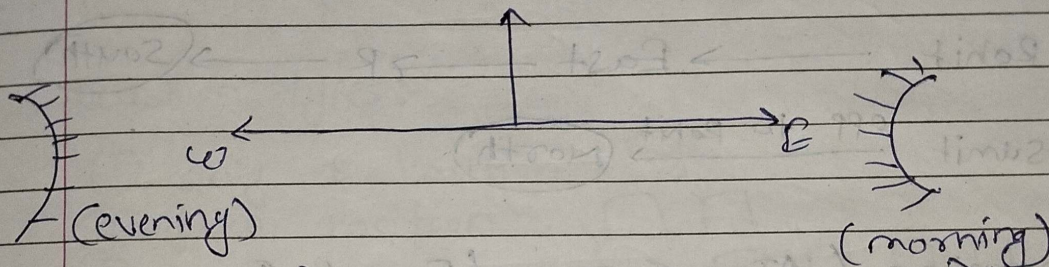
- ③ Head down yoga
 Face $\longrightarrow$ SE $\xrightarrow{\text{opposite}}$ NW
 LH $\longrightarrow$ SW
 RH $\longrightarrow$ NE



[Name] → [E/W] → [R/L]
 Name - जिसके L/R में shadow बता दें

E - morning
 W - Evening
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TYPE-7:- (Shadow)



- morning में shadow west direction में बता दें
 - Evening में shadow ~~west~~ east direction में बता दें।

(1) In a morning Ravi & Rita talking to each other face to face. IF shadow of Ravi falls exactly to the left of Rita. then in which direction Ravi is Facing.

→

[Name] → [E/W] → [R/L]

Name:- जिसके L/R में shadow बता दें।

E - morning

W - Evening.

Rita → East → left → North

Ravi $\xrightarrow[\text{F2F}]{\text{opposite to Rita}}$ South

- (2) morning, sumit & Rohit, Face to Face, shadow of sumit (Right to Rohit). Find direction of sumit facing?

⇒ Rohit → East → R → South
 sumit ^{opp to Rohit} → North

- (3) morning / Ajay & Jaya / Face to Face, shadow of Ajay → (Right to Ajay)

⇒ Ajay → East → R → South
 Jaya → North

- (4) Evening / sumit & Jaya / Face to Face / shadow of Jaya → (Right to sumit)

⇒ sumit → west → Right → North
 Jaya ^{opp to sumit} → South

- (5) Evening / Anil & Vijay / face to Face, shadow of Anil → (Left to vijay)

⇒ Vijay → west → Left → South
 Anil ^{opp to vijay} → North